WhatsApp Simulator & Founder Dashboard

Coding-agent prompt | Internal COD Risk Management application | 31 October 2026

INSTRUCTION TO CODING AGENT: Build a working, polished mock WhatsApp workflow and a distinctive, restrained founder dashboard. Integrate with the existing decision engine; do not rewrite its business rules. Read the original take-home PDF, the decision-engine specification, and orders_sample.csv first. Inspect the current repository before selecting libraries. Present a short implementation plan, then implement and test. Do not invent observed business results.

1. Product overview and scope

This is an internal tool for a D2C haircare brand using Shopify. Roughly 55% of 3,000 monthly orders are COD, with an approximately 28% COD RTO rate versus 4% for prepaid. The system assigns one of four initial actions: ship as-is, ship after WhatsApp confirmation, nudge to prepaid, or hold/cancel. The simulator demonstrates customer messaging without real WhatsApp credentials; the dashboard helps a founder see daily decisions and their estimated rupee impact in seconds.

- Deliver: functional message simulator, founder dashboard, configurable policies, accessible micro-interactions, persistent mock events, clear README, tests, and seeded demo scenarios.
- Do not build: a new storefront, a real payment gateway, a real WhatsApp integration, fake production analytics, or a complex CRM.
- Architecture: reuse existing app stack and APIs. Prefer React/TypeScript for UI if available; use a deterministic backend state machine and SQLite or existing persistence. Avoid duplicating decision logic in the frontend.

2. Agreed business rules (integration contract)

Segment	Initial action / rule
Repeat: trust score >= 2	Ship directly; no WhatsApp.
Repeat: score 0 or 1	Use risk model to decide ship versus confirmation; no prepaid nudge.
Repeat: score -1, or negative with prior successful deliveries	Mandatory WhatsApp confirmation.
Repeat: score <= -2 AND prior successes = 0	Hold/cancel under the agreed rule.
New: below confirmation threshold	Ship directly.
New: above confirmation threshold through 40% RTO	Mandatory WhatsApp confirmation.
New: RTO > 40%	Prepaid nudge: 5% discount capped at INR 50; if declined, offer COD confirmation.

Critical: New customers must not be automatically rejected solely because their modeled RTO risk is high. Unusable address handling belongs to the decision engine: request correction/hold rather than silently treating

an address defect as fraud. Confirmation windows: 48 hours, reminder at 24 hours. The model and confirmation threshold are configurable, not hardcoded in UI.

3. WhatsApp simulator - interaction design

Build a small, believable chat workspace. Left pane: searchable simulated conversations with customer name, order ID, action, time remaining, and unread/status dot. Center: phone-like message thread with restrained WhatsApp-style outgoing/incoming bubbles. Right contextual rail or collapsible drawer: order value, predicted risk, decision reason, status, discount, and event history. On narrow screens use tabs or a drawer; do not squeeze all three columns.

3.1 Message copy - natural, helpful, not robotic

COD confirmation, initial: "Hi {first_name}, thanks for ordering from {brand}! Please confirm your COD order #{order_id} for INR {amount}. Reply YES to confirm or NO to cancel. We will hold it for up to 48 hours."

24-hour reminder: "Quick reminder: we are waiting for confirmation of order #{order_id}. Reply YES to confirm or NO to cancel. Your confirmation window ends in 24 hours."

YES response: "Thanks! Your COD order #{order_id} is confirmed and ready for processing."

NO response: "Understood. Order #{order_id} has been cancelled. If this was a mistake, please contact support."

Prepaid nudge: "Hi {first_name}! You can switch order #{order_id} to prepaid and save INR {discount}. Would you like to pay online? Reply YES for the mock payment link, or NO to keep COD and confirm your order."

Prepaid YES: show a simulated payment link/action, then a separate "Mark payment successful" control. Do not treat a YES as successful payment. On success, transition to prepaid-ready-to-ship; on failure, offer retry or COD confirmation.

Prepaid NO: "No problem - you can continue with COD. Reply YES to confirm this COD order or NO to cancel." This is a NEW confirmation step, not automatic shipment.

3.2 Language and input rules

- Default to clear English; optionally support Hindi/Hinglish templates selected by explicit customer preference or an operator-controlled demo selector. Do not infer language from name, city, or pincode. Keep variable placeholders safe and correctly formatted.
- YES and NO are the only accepted semantic responses; trim whitespace and accept case-insensitive variants "yes"/"no". Other text, emoji, or random input is unrecognized: send one concise clarification, do not change order state.
- For prepaid nudge, YES means interest in prepaid, not payment success; NO means COD fallback. For COD confirmation, YES means confirm and NO means cancel. Store the prompt context so the same reply is interpreted correctly.
- No response at 24h: send reminder and remain pending. No response at 48h: mark held/expired; do not ship. Allow manual review or resend only through explicit operator action.

4. Simulator state machine and controls

State	Event	Next state / side effect
CONFIRM_PENDING	YES	COD_CONFIRMED; eligible to ship
CONFIRM_PENDING	NO	CANCELLED
CONFIRM_PENDING	other	Remain pending; send clarification

State	Event	Next state / side effect
CONFIRM_PENDING	24h elapsed	REMINDER_SENT; remain pending
CONFIRM_PENDING	48h elapsed	HELD_EXPIRED; block shipment
PREPAID_OFFERED	YES	PAYMENT_PENDING; show mock payment action
PREPAID_OFFERED	NO	CONFIRM_PENDING; send COD confirmation
PAYMENT_PENDING	mock success	PREPAID_CONFIRMED; eligible to ship
PAYMENT_PENDING	mock failure	Remain pending; retry or COD fallback

- Expose demo controls: "Reply YES", "Reply NO", "Send other text", "Advance +24h", "Advance to 48h", "Simulate payment success/failure", and "Reset scenario". Label simulated time clearly; never wait 48 real hours.
- Prevent duplicate sends, duplicate transitions, stale callbacks, YES after expiry, confirmation after cancellation, and double shipping. Every transition must be idempotent and logged with timestamp, source, previous state, next state, and reason.
- Show an explicit error or unavailable state when the decision-engine API fails. Do not silently fabricate a successful decision. Provide deterministic seed conversations for every edge case.

5. Founder dashboard - design direction

Design concept: "Editorial operations ledger". Avoid the generic AI dashboard: no giant gradient hero, no glowing cards, no gratuitous doughnut charts, no 12 KPI tiles, no dense rainbow badges. Use a warm off-white canvas, ink-navy typography, subtle ruled dividers, one muted teal accent, restrained amber for review, and red only for genuine losses/alerts. Typography should feel like a polished financial publication: strong numeric hierarchy, small uppercase section labels used sparingly, generous whitespace, consistent alignment, tabular numbers, and quiet micro-interactions.

Visual goal: within 5 seconds the founder can answer "What happened yesterday? Did it save money? Where should I look?" Provide useful depth on hover, click, or disclosure, not by cramming explanations into the main canvas.

5.1 Above-the-fold information hierarchy

- Top bar: "COD operations / Yesterday", exact date and timezone, a date selector, last-updated timestamp, and compact scenario/assumption indicator. One primary action: "Review exceptions".
- Hero figure: Net estimated INR impact as the visual anchor, accompanied by baseline comparison and an uncertainty label such as "Scenario estimate". Small supporting values: estimated RTO losses avoided, margin lost on genuine cancelled/held orders, and intervention costs. Do not use a fabricated upward arrow.
- Four-bucket decision strip: Ship directly / WhatsApp confirmation / Prepaid nudge / Hold-cancel with counts and percentage of yesterday's orders. Compact segmented horizontal bar; show exact counts in adjacent text for accessibility.
- One focused insight panel: "What changed" with the top grounded explanation, e.g. "More orders entered confirmation than yesterday" only when a comparison exists. Otherwise state "No comparison available".
- Exceptions queue: at most five rows by default, prioritized by urgency: 48h expired, ambiguous address, high-value hold, failed payment, manual review. Columns: order, reason, next action, age. "View all" opens a full list.

5.2 Hover, focus, and progressive disclosure

- Every metric label has an accessible info trigger (hover on desktop, tap/click and keyboard focus everywhere). Tooltip/popover explains definition, formula, date window, denominator, assumptions, and whether value is

observed or estimated. Keep tooltips 2-4 short lines; add "View calculation" for longer details.

- Clicking any bucket filters the orders table and opens a side drawer with a decision breakdown, sample reasons, actual/predicted RTO where available, and the policy that assigned the action. Do not navigate away unnecessarily.
- Hovering a chart segment shows count, percentage, and a meaningful short note. Provide keyboard equivalent and a text summary. No information should be accessible by hover only.
- Use skeleton loading, purposeful empty states, error/retry states, and responsive layouts. Motion should be subtle, under 200 ms, and respect reduced-motion preferences.

6. Dashboard metric definitions - do not mislead

Decision counts: count each order once by its initial decision for the four-bucket overview. Separately show its latest operational state (confirmed, converted, held, cancelled). Do not double-count a prepaid nudge that later falls back to COD confirmation.

Baseline: estimate what would have happened if all eligible COD orders shipped without intervention. Historical delivered/RTO labels can be used for retrospective backtests, but not to claim actual avoided outcomes in a live demo.

Estimated avoided RTO loss: sum modeled prevented-RTO probability x INR 169.40 (65 forward + 65 return + 25 packaging + 8% x 180 damage). Avoided shipping costs apply only when an order is stopped before dispatch. If converting to prepaid, model its own RTO and margin outcomes separately.

Estimated lost genuine margin: sum probability an intervention blocks a would-have-delivered order x 0.62 x order value; make the assumptions explicit. Discounts reduce revenue and contribution, not the product cost; payment fees and message costs are configurable assumptions.

Net INR impact: expected contribution under intervention policy MINUS expected contribution under ship-all-COD baseline. Include discount, messaging, and assumed processing fees. Never present scenario projections as realized cash savings.

Founder confidence: show observed values distinctly from estimated scenario outputs. Add a compact "Assumptions" drawer with prepaid conversion rate, WhatsApp RTO prevention, genuine nonresponse, prepaid RTO, discount and messaging cost. Changing these recalculates projections; label changes as simulations.

7. Minimal API and data contract

Endpoint / object	Required behavior
GET /dashboard?date=YYYY-MM-DD	Returns daily initial bucket counts, net estimate, components, assumption set, last updated, exceptions.
GET /orders?decision=&status;=	Paginated orders with initial decision, latest state, risk, reason, timestamp.
GET /simulator/conversations	List of seeded and created conversations; search/filter.
POST /simulator/{id}/reply	Body: {text}; resolve based on current prompt state; return messages + state.
POST /simulator/{id}/advance	Body: {hours}; deterministic mock clock, reminder/expiry transitions.
POST /simulator/{id}/payment	Body: {outcome: success|failure}; mock only.
GET/PUT /settings	Editable business assumptions and thresholds, with validation and audit trail.

Use stable order IDs and event IDs; persist events and initial decisions. Monetary fields should be integer paise or Decimal, never binary float for financial totals. Dates use ISO 8601 with explicit timezone; display in Asia/Kolkata. Validate API input and return meaningful HTTP errors. The UI should work from seeded data without a live LLM or WhatsApp account.

8. Acceptance tests and demo walkthrough

- Scenario 1: new medium-risk COD order -> confirmation message -> YES -> confirmed -> dashboard operational status updates.
- Scenario 2: confirmation -> random reply -> clarification -> +24h reminder -> +48h hold; ensure no shipment.
- Scenario 3: new >40% risk -> 5% prepaid offer capped INR 50 -> YES -> mock payment success -> prepaid confirmed; ensure discount accounted for.
- Scenario 4: prepaid NO -> COD confirmation -> YES -> confirmed; do not interpret the first NO as order cancellation.
- Scenario 5: repeat score >=2 -> direct ship without WhatsApp; repeat score -1 -> mandatory confirmation; score <=-2 with zero successes -> hold/cancel.
- Scenario 6: dashboard bucket totals reconcile to eligible order count; changing assumptions updates estimated net impact, not observed counts.
- Scenario 7: keyboard navigation, screen reader labels, touch popovers, narrow viewport, no hover-only content, graceful API error.

9. Implementation order and handoff

Phase A: inspect existing engine and data contracts; write an integration map and UI wireframe. Phase B: implement simulator state machine, seeded scenarios, persistence and tests. Phase C: build the editorial dashboard with real derived metrics and progressive disclosure. Phase D: add responsive polish, assumptions panel, accessibility and end-to-end walkthrough. Phase E: README with setup under 10 minutes, screenshots, tests, known limitations, and explicit assumptions.

Final instruction: Do not substitute attractive but fake metrics for correct calculations. If a metric cannot be computed from available data, mark it "Unavailable" or "Scenario estimate" and document the missing evidence. Prioritize a coherent working flow, restrained visual quality, and explainable business decisions over breadth.